Does yield hide flood losses? Harvested area, flood timing and Boro rice in Bangladesh's 64 districts, 2011–2025

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Abstract

Flood losses to crops are often measured through yield, but official yield is estimated on harvested fields, so a destroyed field may leave no trace in it. We tested this for Boro rice in Bangladesh, using a validated panel of 64 districts for 2011–2025, flood fractions from Sentinel-1 and JRC, and an analysis plan fixed in advance. In the 2017 flash flood, harvested area in the seven haor districts fell by 31 % relative to other districts, while yield changed by −0.9 %. In Sunamganj, area fell by 72 % and yield by 0.8 %. A satellite crop-loss measure and published damage figures support these losses. The general result was only suggestive under the plan's rule: the area effect per 10 points of flood was −21.6 % with Sentinel-1 and −4.7 % with JRC, and it disappeared without 2016 and 2017. The 2016 and 2022 floods left no fall in area. An exploratory analysis suggests that timing matters, but rests on 2017. Yield alone can therefore miss the largest losses, so damage assessment should also track harvested area and production.

Keywords: flood loss; flash flood; Boro rice; harvested area; crop yield statistics; Bangladesh

Software and data availability

- Code, notebooks and outputs: https://github.com/TasMumu/bd-boro-flood-risk [authors: make the repository public, archive a release and give its DOI here before submission]
- Data: the panel of Mumu and Karim (2026), archived at https://doi.org/10.5281/zenodo.23795046. The Earth Engine extracts used here are included with the code
- Software: Python 3.13 with the packages listed in the code repository, and Google Earth Engine for the new extracts

1. Introduction

Assessing what a flood costs a farming region depends on what is measured. Most research on weather and crops has focused on yield (Cui, 2020; Wei and Davis, 2022), and yield is the natural outcome when a flood reduces grain per hectare. But a flood can also destroy a crop outright. If official yield is measured

only on fields that were harvested, a destroyed field does not lower the yield estimate. It simply leaves the harvested area. Disaster assessments and research that look at yield alone could then miss the losses that matter most.

The evidence on this point is mixed, and the answer appears to depend on the setting. Globally, drought reduced cereal production through both area and yield, extreme heat mainly through yield, and no effect of floods could be identified in national data (Lesk et al., 2016). A recent global study reports that submergence for more than a week reduces annual rice yield by 4.3 % (Li et al., 2025). In the United States, harvested ratios fall under extreme heat (Cui, 2020), and area-related disruptions explain at least as much maize production instability as yield (Wei and Davis, 2022). In Kansas wheat, however, losses came from yield and not from abandonment (Obembe et al., 2021). For rice, typhoons reduced provincial production in a Philippine panel (Blanc and Strobl, 2016).

Boro rice in Bangladesh is a useful test. It is the country's main rice crop, grown in the dry season and harvested from April to May. In the haor wetlands of the north-east, early flash floods can destroy the standing crop shortly before harvest (Kamruzzaman and Shaw, 2018; Rashid and Yasmeen, 2018), and the 2017 flood alone was reported to have cost 0.88 million tonnes of Boro (Hossain et al., 2023). The Bangladesh Bureau of Statistics (BBS) estimates yield as production per harvested hectare, from crop cuts on harvested fields (Bangladesh Bureau of Statistics, 2026), so the reporting explanation can be tested directly: if destroyed fields leave no trace in yield, a flood should appear in area and production but not in yield.

We asked five questions. Do flash floods reduce Boro production through harvested area rather than yield? Do monsoon floods, which come before planting, act differently? Are flood effects concentrated in the haor? Why does the vegetation index NDVI not explain yield within a district? And where is risk concentrated, and how stable is that ranking?

The analysis makes three contributions. It uses a validated panel of all 64 districts, with values printed by BBS and no modelled yields (Mumu and Karim, 2026). It checks official statistics against evidence independent of them, namely two flood records, a satellite measure of crop loss, and published damage figures. And its questions, models and pass rules were fixed before the models were run, so that a weak result is reported as weak. That turned out to matter: the area effect is clear for the 2017 flood, but the evidence for a general rule is only suggestive.

2. Background

2.1 Flash floods and Boro in the haor

The haor basin is a low-lying wetland in north-eastern Bangladesh, where Boro is often the only crop of the year (Kamruzzaman and Shaw, 2018). Two hazards dominate: cold at the reproductive stage and flash flood at maturity (Rashid and Yasmeen, 2018). Floods are typically described as arriving in the first two weeks of April (Hossain et al., 2023), and the 2017 flood began on 27 March (Ahmed et al., 2017). Timing is therefore central, because the same water can destroy a standing crop or arrive after the harvest. Evidence on these losses is mostly from single events or small areas: satellite mapping of the 2017 damage (Ahmed et al., 2017), household surveys (Dey et al., 2021; Ahmed, 2024) and local studies (Khan et al., 2012; Mondal et al., 2021). The haor is also an adaptation setting, where short-duration varieties and earlier sowing are recommended (Hossain et al., 2023) and farmers report barriers to adapting (Baishakhy et al., 2023).

2.2 Floods and rice in Bangladesh

In Bangladesh, floods are not simply harmful. Districts more prone to flooding had higher cultivated area and productivity in 1978–2000, with yield falling in extreme floods and rising in normal floods and in the months after (Banerjee, 2010). Boro production rose after the 1988 and 1998 floods, helped by residual water, while the Aman crop fell through both area and yield (Asada et al., 2005). A district panel found meaningful flood effects on Boro production but no meaningful yield results (Hassan, 2019). Studies of climate and rice mostly examine yield, using regions, stations or national series (Sarker et al., 2014; Hasan et al., 2016; Rahman et al., 2017). One study covers both yield and cropping area, using national time series and climate variables (Amin et al., 2015). Rainfall and self-reports are weak proxies for flood exposure in Bangladesh, which supports the use of satellite data (Guiteras et al., 2015).

2.3 The gap

Table 1 compares the closest studies with this one. We found no study that uses a multi-year district panel with satellite flood data to split Boro flash-flood losses into harvested area and yield. This rests on a documented search of 1,383 records and on abstracts rather than full texts. One recent paper on monsoon floods and crop production in Bangladesh (Rahman et al., 2025) could be read only in part.

Table 1. Closest studies compared with this paper.

Study	Flood type	Unit and period	Satellite flood data	Area and yield separated	Haor focus
Banerjee (2010)	Monsoon	Districts, 1978–2000	Not stated	Partly	No
Asada et al. (2005)	Severe monsoon	National, long term	No	Yes, descriptive	No
Hassan (2019)	Flood damage indicators	Districts	Not stated	No	No
Amin et al. (2015)	None, climate only	National, 1972–2010	No	Yes	No
Ahmed et al. (2017)	Flash, 2017 only	North-east, one event	Yes	Damaged area only	Yes
This paper	Flash and monsoon	64 districts, 2011–2025	Yes, two records	Yes	Yes

3. Data

3.1 Panel

The analysis uses the Gold panel of Mumu and Karim (2026): 960 district-years for all 64 districts and harvest years 2011–2025, where harvest year t is the crop sown in November of t−1 and harvested in April–May of t. Boro harvested area, yield and production are values printed in the BBS yearbooks, each with its edition and page, and no value is modelled or filled. Weather comes from ERA5-Land (Muñoz-Sabater et al., 2021). Flood fractions come from the monthly surface water record of the Joint Research Centre (JRC) (Pekel et al., 2016) and from Sentinel-1 radar (Torres et al., 2012), and are kept as separate records. Table 2 lists the sources. The seven haor districts are Sunamganj, Habiganj, Netrokona, Kishoreganj, Sylhet, Moulvibazar and Brahmanbaria, a list fixed before the models were run. They produced 17.5 % of Boro over the period.

Table 2. Data sources.

Variable	Source	Resolution	Period
Boro area, yield, production	BBS yearbooks, through the Paper 1 panel	64 districts	2011–2025
Flood, Sentinel-1	COPERNICUS/S1_GRD, VV, −18 dB	Analysed at 60 m	Harvest years 2015–2025
Flood, JRC	Global Surface Water v1.4, monthly	30 m	Harvest years 2011–2021
Boro mask, crop loss, NDVI	MODIS MOD13Q1 v061, WorldCover v200 cropland	250 m	2011–2025
Weather	ERA5-Land daily aggregates	0.1°	2011–2025

Flash flood is the share of district area under water in April or May of year t and not in January–March of t. Monsoon flood is the share under water in June–October of t−1 and not in January–March of t−1. JRC

ends in December 2021 and Sentinel-1 begins in October 2014, so the main models use harvest years 2011–2021 for JRC and 2016–2025 for Sentinel-1.

3.2 New variables

New variables were computed in Google Earth Engine (Gorelick et al., 2017). Every extract was tested for completeness, and no value was filled.

Boro mask. Published Boro maps cover only 2017 (Singha et al., 2019) or 2014–2018 (Shew and Ghosh, 2019), so we built a consistent mask for the whole period, following earlier MODIS mapping of Boro (Mosleh and Hassan, 2014). Using the normalised difference vegetation index (NDVI) from MODIS (Didan, 2021) on WorldCover cropland (Zanaga et al., 2022), a pixel is counted as Boro land in year t if its peak NDVI from February to mid-March reaches 0.50 and has risen by at least 0.20 since December–January. These images all end before the flood window, so the mask describes the crop before any loss. Area changes little for thresholds between 0.40 and 0.50. A stable mask, covering land that meets the rule in at least five of the 15 years, is used for flood fractions on Boro land. It covers 4.70 million ha, against about 4.8 million ha harvested in BBS statistics, and agrees with BBS area between districts ($r = 0.91$). The yearly mask is noisier (Table A.2) and is used only to define the land on which crop loss is measured.

Satellite crop loss. Following the logic of the Disaster Vegetation Damage Index (Rahman et al., 2021), we took the change in NDVI from the pre-flood peak to the two composites starting in April, and subtracted the same pixel's median change over all years. Loss is the share of Boro land with an anomaly below −0.20. This measure uses no BBS data.

Flood timing. The flash-flood window was split into early and late parts against a January–February baseline, so that water from a March flood is not treated as dry-season water. For Sentinel-1, early is 15 March to 15 April and late is 16 April to 15 May, each from two half-month medians. JRC is monthly, so early is March–April and late is May.

Other variables. We also computed flood fractions at water thresholds of −16 and −20 dB, Sentinel-1 scene counts, hot days above 35 °C from mid-March to April (Jagadish et al., 2007), and days below 20 °C from mid-February to March. The cold threshold is provisional. Adaptive capacity is not covered, because no published district series was assembled, and nothing was estimated in its place.

3.3 Data checks

All 18 checks passed. The new code recomputed the published flood fractions of the panel, and the largest difference was below 10^{-9}, so the variants are computed by verified code. The panel's checksum matches the archived version.

4. Methods

4.1 Analysis plan

Questions, models, pass rules and the bound for a negligible yield effect were fixed in a written plan before any new model was run. The basic models in the framework paper (Mumu and Karim, 2026) had been seen and count as exploratory. Two changes were made before the outcome models were run, based on flood and mask data and BBS area between districts: flood fractions on rice land use a stable mask built here, because the yearly mask proved noisy and published rice maps do not cover the period; and Sentinel-1 early and late windows use half-month medians, because a single monthly median diluted a flood arriving midway. After the planned results were seen, we added the analyses in Section 5.6 and one extra synthetic-control test. These are labelled as exploratory, and they do not change the verdict.

4.2 Flood models

For district i in harvest year t, each outcome Y (harvested area, yield or production) is modelled as

$$\ln Y_{it} = \beta F_{it} + \gamma M_{it} + \mathbf{x}_{it}'\boldsymbol{\delta} + \alpha_i + \tau_t + \varepsilon_{it},$$

where F is flash flood and M monsoon flood, both per 10 points of district area, **x** holds mean temperature, precipitation and hot days, and α and τ are district and year effects. Since production is area times yield, the area and yield effects must add up to the production effect; the largest gap was below 0.0001 log points. Effects are reported in per cent. This is the standard panel approach to weather effects (Hsiang, 2016; Blanc and Schlenker, 2017; Auffhammer et al., 2013). Results are estimates under the stated assumptions: floods are not randomly assigned, and we avoid causal language.

4.3 Three designs for flash floods

Flash floods were studied in three ways that must point the same way: the dose-response model above; a year-by-year comparison of haor and other districts relative to 2015, the last year before the two largest floods; and synthetic control (Abadie, 2021) for the five most flooded districts in 2017, where a weighted mix of non-haor districts matches each district's path in 2011–2015.

4.4 Uncertainty

Few districts are treated and they are atypical, a setting in which clustered errors and the wild bootstrap can fail, and randomisation inference has limits (Conley and Taber, 2011; MacKinnon and Webb, 2017, 2018, 2020). So no single method was relied on. Every main estimate is reported with errors clustered by district, Driscoll–Kraay errors (Driscoll and Kraay, 1998), Conley spatial errors (Conley, 1999) with a 100 km cut-off and two years of serial correlation, and randomisation inference that reassigns whole flood years across the panel, keeping each flood's real spatial pattern. With T years, one event alone cannot reach a p-value below 1/T, about 0.10 here, and two events below about 0.01.

4.5 How results were judged

A result is called **supported** only if it has the same sign and similar size under all three designs, $p < 0.05$ under all four uncertainty methods in both flood records, the same direction in both records, and passes every required check. If the significance rule holds in one record it is **suggestive**, and otherwise **not supported**. Leave-out checks pass with the same sign and at least half the main size. Placebos pass if not significant at 5 % under clustered and Driscoll–Kraay errors. Satellite crop loss passes if flood raises it and it lowers BBS area, each with $p < 0.05$ under Driscoll–Kraay errors. Measurement checks pass with the same sign and $p < 0.05$ under Driscoll–Kraay errors. Synthetic control passes if the 2017 gap is negative and its size relative to the pre-flood fit ranks in the top 5 % of placebo districts.

A yield effect counts as negligible only if its 95 % interval lies within ±2 % per 10 points under every interval method. An insignificant coefficient alone is not evidence of no effect.

4.6 Index and spatial analysis

The risk index combines exposure (flash and monsoon flood on Boro land, heat at flowering and cold at the reproductive stage) with sensitivity (production instability, worst fall in area, low yield). In line with the handbook on composite indicators (OECD and European Commission Joint Research Centre, 2008) and uncertainty analyses of such indices (Saisana et al., 2005; Tate, 2012), we rebuilt it under 6,018 combinations of three normalisations, two aggregations and four weighting schemes, including 1,000 random draws of weights. Global Moran's I (Moran, 1950) and LISA (Anselin, 1995) were run under four neighbour definitions, with false discovery rate correction (Benjamini and Hochberg, 1995). Monsoon-flood and haor-interaction tests were corrected the same way as secondary tests.

5. Results

5.1 Few events carry the flash-flood estimates

Flash floods are concentrated in a few places and years (Fig. 1, Fig. 2). Sentinel-1 has 704 district-years but only about 11 effective observations (Table 3). Seven districts carry nearly all the variation, and two years carry most of it, so the flash-flood analysis is really a study of a few events. Monsoon floods are spread much more widely. Sentinel-1 also has a coverage break: scenes per haor district more than double after 2016 (Fig. 2b), which matters because 2016 is a key flood year.

Fig. 1. Flash flood and satellite crop loss. Panel (a) shows mean flash flood on Boro land. Panel (b) shows the share of Boro land with a sharp fall in greenness in 2017.

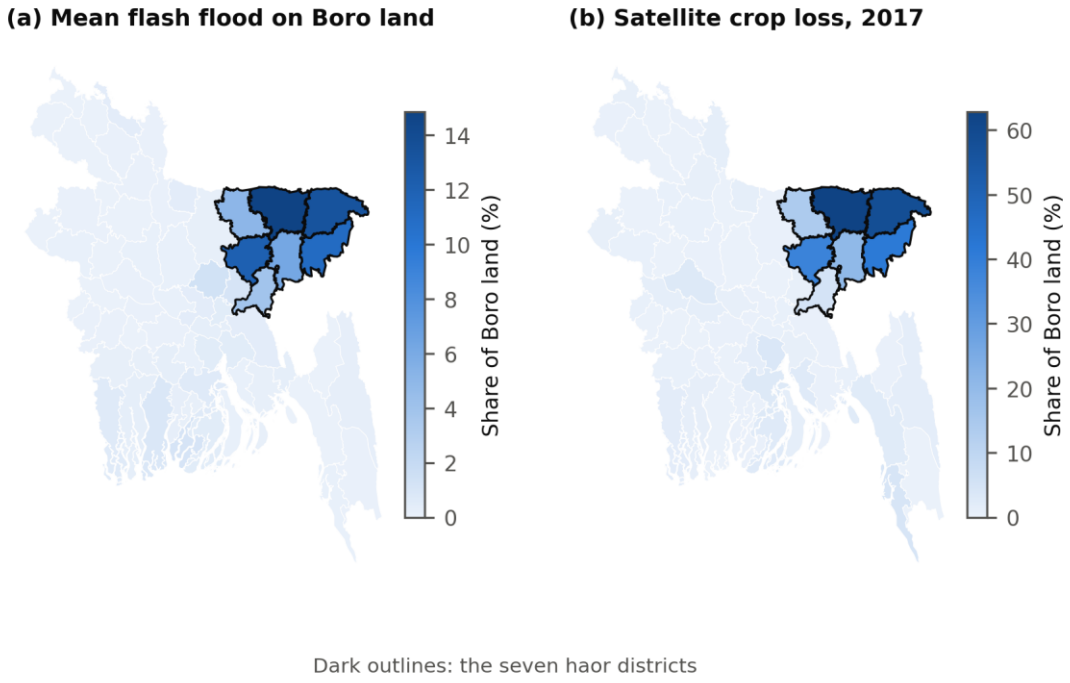
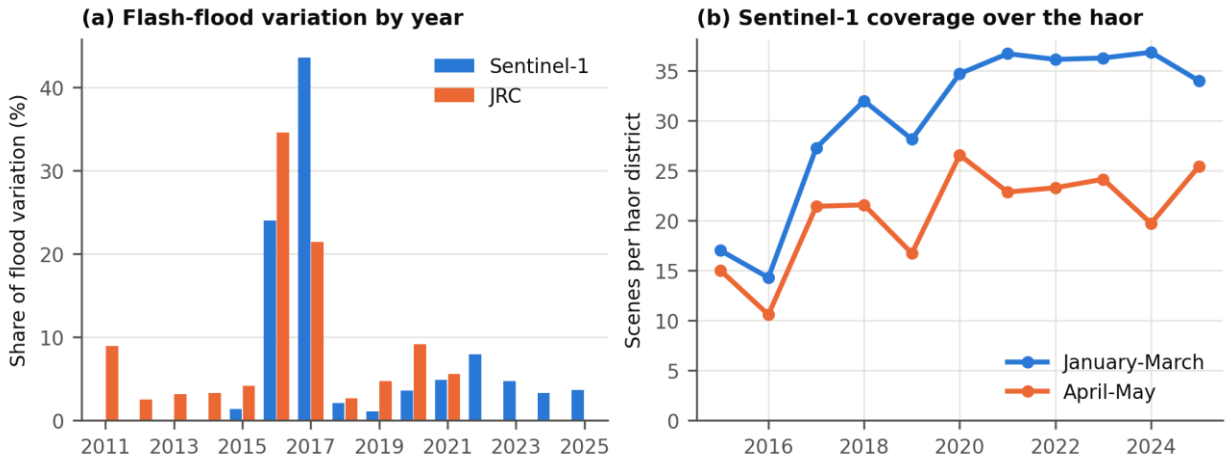


Fig. 2. How few events carry the estimates. Panel (a) shows each year's share of flash-flood variation after removing district and year means. Panel (b) shows Sentinel-1 scenes per haor district.



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198 Table 3. Where the flood variation comes from.

Flood variable	District-years	Effective observations	From the seven haor districts (%)	From 2016 and 2017 (%)
Flash flood, Sentinel-1	704	11	88	68
Flash flood, JRC	704	23	90	56
Monsoon flood, Sentinel-1	640	52	47	13
Monsoon flood, JRC	768	103	41	19

199 ## 5.2 Main flood models

200 With Sentinel-1, 10 more points of flash flood went with 21.6 % less harvested area and 21.7 % less
201 production, significant under all four methods, and with no detectable change in yield (−0.2 %) (Table 4).
202 With JRC the area effect was much smaller (−4.7 %) and not significant under any method. Yield rose
203 slightly with JRC (1.1 %), significant under Driscoll–Kraay errors and randomisation inference only.

204 Table 4. Flash-flood effects per 10 points of district area, with p-values from four methods.

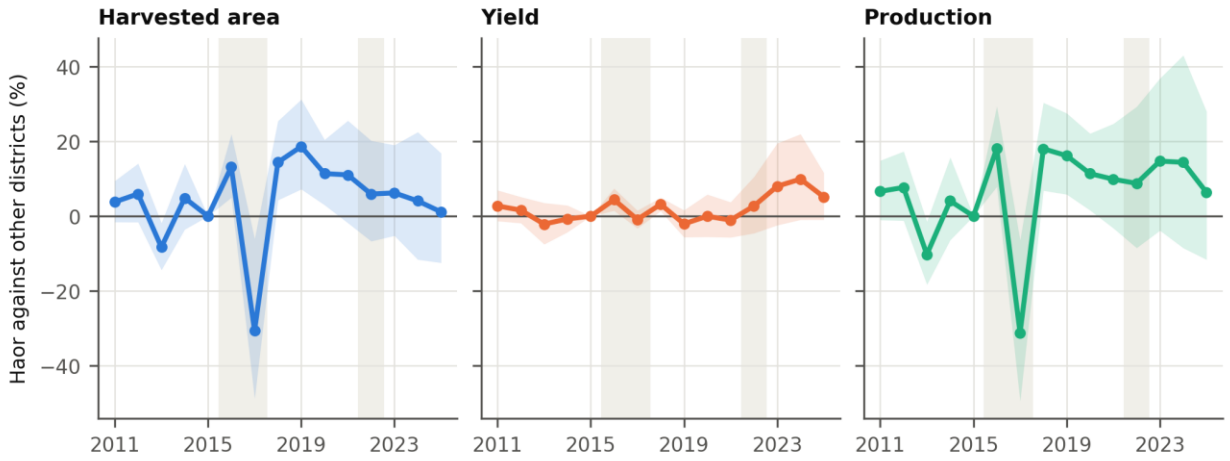
Flood record	Outcome	Effect per 10 points (%)	Clustered p	Driscoll–Kraay p	Conley p	Randomisation p	N
Sentinel-1	Harvested area	−21.6	0.021	< 0.001	0.007	0.002	640
Sentinel-1	Yield	−0.2	0.862	0.878	0.871	0.891	640
Sentinel-1	Production	−21.7	0.027	< 0.001	0.010	0.004	640
JRC	Harvested area	−4.7	0.080	0.140	0.114	0.238	704
JRC	Yield	1.1	0.098	0.022	0.051	0.013	704
JRC	Production	−3.7	0.167	0.215	0.220	0.369	704

The yield interval was narrow but just exceeded the ±2 % bound: −2.3 to 2.0 % for Sentinel-1 under Driscoll–Kraay errors, and up to 2.4 % for JRC under clustered errors. So the planned test for a negligible yield effect was not met. Yield effects were small, and no method gave a sizeable negative one.

5.3 The 2017 flood

The year-by-year comparison shows why the two records differ (Fig. 3). In 2017, haor districts fell 30.6 % in area and 31.2 % in production relative to other districts and to 2015, while yield changed by −0.9 %. Area recovered at once, to 14.4 % in 2018. The other large floods left no such mark: haor area was 13.1 % in 2016 and 5.9 % in 2022. Differences before the floods reach 8.3 %, so smaller movements should not be over-read.

Fig. 3. Haor districts against other districts, relative to 2015, on one shared scale. Flood years are shaded. Intervals are clustered by district and can be too narrow with few treated districts.



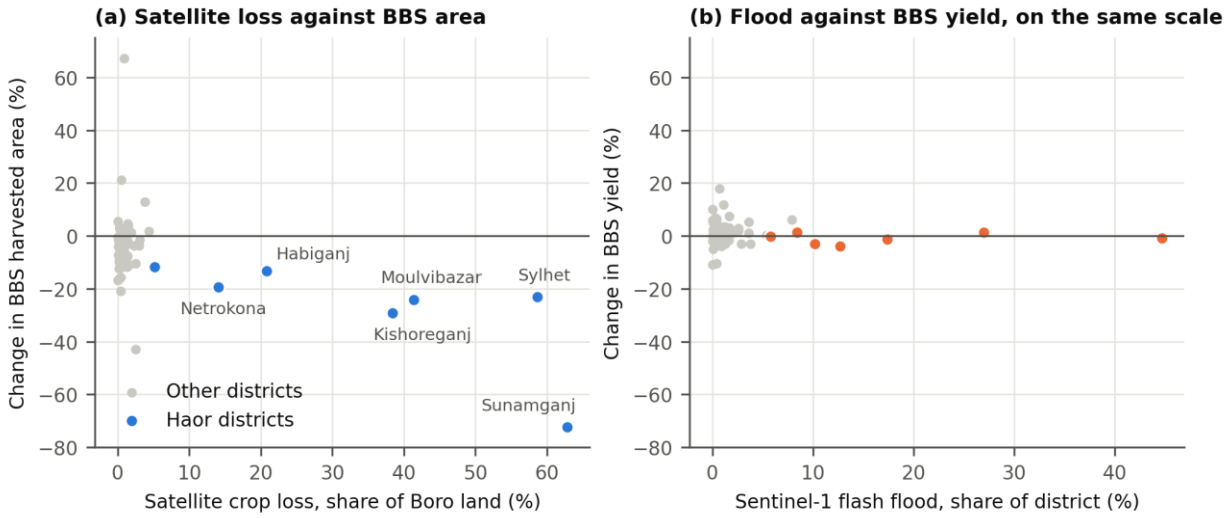
Sunamganj shows the pattern most clearly: 189 thousand ha harvested in 2016, 54 thousand ha in 2017 and 201 thousand ha in 2018. Against the mean of 2015 and 2018, its area fell by 72.4 % and its yield by 0.8 % (Table 5, Fig. 4).

Table 5. The 2017 flood in the most flooded districts. BBS changes are from the mean of 2015 and 2018.

District	Sentinel-1 flood (%)	JRC flood (%)	Satellite crop loss (%)	BBS area (%)	BBS yield (%)	BBS production (%)
Sunamganj	44.7	37.2	62.8	−72.4	−0.8	−72.6
Kishoreganj	27.0	30.0	38.4	−29.2	1.3	−28.2
Netrokona	17.4	16.2	14.1	−19.4	−1.3	−20.5
Sylhet	12.7	26.7	58.6	−23.1	−3.8	−26.1

District	Sentinel-1 flood (%)	JRC flood (%)	Satellite crop loss (%)	BBS area (%)	BBS yield (%)	BBS production (%)
Habiganj	10.2	14.1	20.8	−13.2	−3.1	−15.9
Moulvibazar	8.4	9.5	41.4	−24.1	1.3	−23.1
Lalmonirhat	7.9	0.4	0.3	−3.0	6.2	2.9
Brahmanbaria	5.8	5.7	5.2	−11.8	−0.1	−12.0

Fig. 4. The 2017 flood across all 64 districts. Panel (a) sets satellite crop loss against BBS harvested area. Panel (b) sets the flood against BBS yield, on the same scale.



Synthetic control gave the same pattern. The estimated 2017 area gaps were −70.4 % in Sunamganj, −29.3 % in Kishoreganj, −17.4 % in Sylhet, −13.2 % in Netrokona and −8.5 % in Habiganj, while yield gaps stayed within 3.9 % (Table 6). The planned test was not met: no gap ranked in the top 5 % relative to its pre-flood fit. With five years to match and many donors, the pre-flood fit is almost perfect, so this ratio is unstable. An exploratory test on the gap's size alone ranks Sunamganj at p = 0.017 and Kishoreganj at p = 0.052, and the other three above 0.1.

Table 6. Synthetic control for the five most flooded districts. Placebo rank p is the planned test.

District	Outcome	Gap 2016 (%)	Gap 2017 (%)	Gap 2018 (%)	Placebo rank p
Sunamganj	Harvested area	0.8	−70.4	3.5	0.103
Kishoreganj	Harvested area	1.6	−29.3	−9.1	0.103
Netrokona	Harvested area	3.8	−13.2	2.1	0.776
Sylhet	Harvested area	−3.3	−17.4	5.1	0.121
Habiganj	Harvested area	1.3	−8.5	−1.8	0.603
Sunamganj	Yield	1.4	−1.1	1.1	0.241

District	Outcome	Gap 2016 (%)	Gap 2017 (%)	Gap 2018 (%)	Placebo rank p
Kishoreganj	Yield	0.5	1.7	3.6	0.431
Netrokona	Yield	1.9	−2.3	2.5	0.483
Sylhet	Yield	−1.6	−3.9	2.1	0.759
Habiganj	Yield	0.5	−0.1	0.7	0.759

5.4 Evidence independent of BBS

The satellite measure uses no BBS data, and it passed its planned test. Each 10 points of flood raised satellite crop loss by 7.5 points with Sentinel-1 and 6.2 points with JRC, and satellite loss lowered BBS area (Driscoll–Kraay p = 0.041) but not yield (p = 0.226). In 2017, satellite loss reached 63 % of Boro land in Sunamganj and 59 % in Sylhet, and stayed below 4 % outside the haor (Fig. 1b, Fig. 4a).

Published damage figures are also of the same order. Against the mean of 2015 and 2018, BBS shows a production loss of 1.08 million t in the seven haor districts in 2017, compared with 0.88 million t reported for the haor regions (Hossain et al., 2023). For Sunamganj, BBS implies 142,482 ha and 520,428 t lost, compared with 149,224 ha affected and 393,855 t damaged in a published estimate (Mondal et al., 2021).

5.5 Checks and verdict

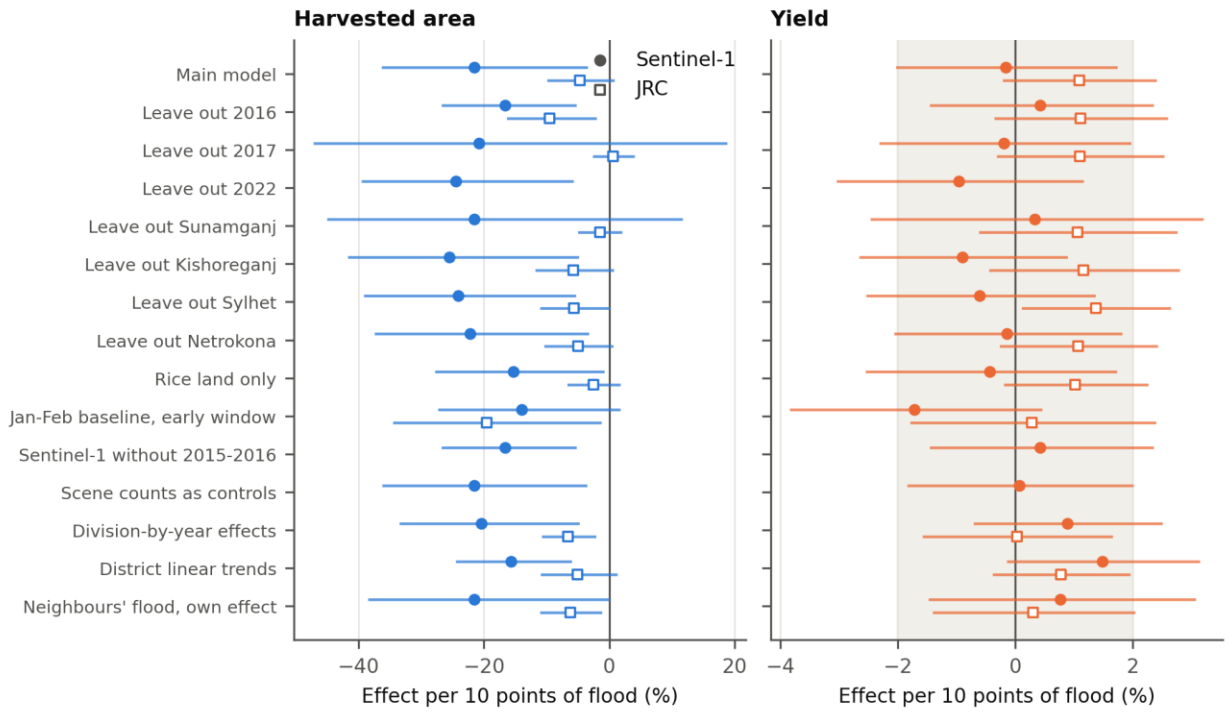
The required checks gave a mixed picture (Table 7, Fig. 5). All leave-out checks passed for Sentinel-1, and the flood-after-harvest placebo passed in both records. Next year's flood, however, failed for Sentinel-1 (11.6 %, clustered p = 0.033), possibly because haor area was high in 2016, just before the 2017 flood. For JRC, the effect disappeared without 2017 (0.6 %) or Sunamganj (−1.5 %), and was not significant on rice land only. Most importantly, without both 2016 and 2017 no effect remained in either record (0.1 % and −0.1 %). Flood size also mattered: only the largest group, 20 % or more of a district, showed clear losses (−31.9 % with Sentinel-1, from 5 district-years, and −15.7 % with JRC, from 8; Fig. 6).

Table 7. Required checks of the area effect. Effects are per 10 points of flood.

Check	Flood record	Area effect (%)	Clustered p	Driscoll–Kraay p	Result
Leave out 2016	Sentinel-1	−16.7	0.005	< 0.001	Pass
Leave out 2016	JRC	−9.5	0.011	0.013	Pass
Leave out 2017	Sentinel-1	−20.8	0.258	0.004	Pass
Leave out 2017	JRC	0.6	0.688	0.752	Fail
Leave out 2022	Sentinel-1	−24.4	0.012	< 0.001	Pass

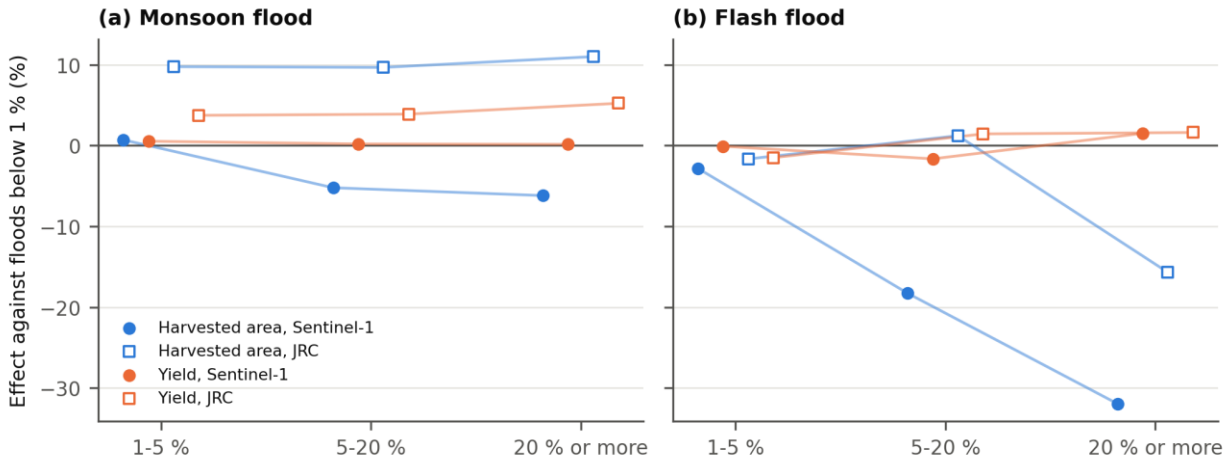
Check	Flood record	Area effect (%)	Clustered p	Driscoll–Kraay p	Result
Leave out Sunamganj	Sentinel-1	−21.6	0.176	0.009	Pass
Leave out Sunamganj	JRC	−1.5	0.379	0.533	Fail
Leave out Kishoreganj	Sentinel-1	−25.5	0.018	< 0.001	Pass
Leave out Kishoreganj	JRC	−5.7	0.073	0.151	Pass
Leave out Sylhet	Sentinel-1	−24.1	0.014	< 0.001	Pass
Leave out Sylhet	JRC	−5.6	0.047	0.101	Pass
Leave out Netrokona	Sentinel-1	−22.2	0.023	< 0.001	Pass
Leave out Netrokona	JRC	−5.0	0.070	0.122	Pass
Next year’s flash flood (placebo)	Sentinel-1	11.6	0.033	0.225	Fail
Next year’s flash flood (placebo)	JRC	0.6	0.678	0.820	Pass
Flood after harvest (placebo)	Sentinel-1	0.9	0.758	0.756	Pass
Flood after harvest (placebo)	JRC	2.3	0.328	0.125	Pass
Rice land only	Sentinel-1	−15.3	0.037	0.001	Pass
Rice land only	JRC	−2.5	0.220	0.233	Fail
January–February baseline, early window	Sentinel-1	−14.0	0.077	< 0.001	Pass
January–February baseline, early window	JRC	−19.5	0.036	< 0.001	Pass
Sentinel-1 without 2015–2016	Sentinel-1	−16.7	0.005	< 0.001	Pass
Scene counts as controls	Sentinel-1	−21.5	0.020	< 0.001	Pass

Fig. 5. How the flash-flood estimates hold up. Each row reruns the main model with one change. Intervals are 95 %, clustered by district. The shaded band marks ±2 %.



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253 Fig. 6. Normal and extreme floods, against the group below 1 %. Bins with few district-years are
254 uncertain.



255

256 Under the rule fixed in advance, the answer to the main question is **suggestive**, not supported. The
257 significance rule held for Sentinel-1 but not for JRC, 19 of 23 required checks passed, and the planned
258 synthetic-control test failed. Direction and design agreed throughout. For Q3, concentration in the haor
259 could not be tested properly: outside it, Sentinel-1 flood reached 5 % in only 1.8 % of district-years, so
260 any slope there is an extrapolation, and no haor interaction survived false discovery rate correction.

5.6 Flood timing (exploratory)

Why did 2017 stand out? This analysis was added after the planned results were seen. Table 8 suggests that timing is the reason. In 2017, more water arrived early, while in 2016 most Sentinel-1 water came late. The planned early-window check had already brought the two records close together: −14.0 % with Sentinel-1 and −19.5 % with JRC per 10 points (Table 9). This is consistent with the JRC estimate in Table 4 being weak because its window mixes water before and after harvest.

This reading has clear limits. Without 2017 the early-window effect disappears or reverses in both records, so it rests on that one event. And satellite loss in 2016 (16.2 %) did not match BBS, where area rose. NDVI also falls when fields are harvested early, so satellite loss cannot separate harvest from destruction.

Table 8. Flood timing in the haor districts. Flood and loss are mean shares; BBS area is from the year-by-year model.

Harvest year	Sentinel-1 early (%)	Sentinel-1 late (%)	JRC early (%)	JRC late (%)	Satellite crop loss (%)	BBS area against 2015 (%)
2015	0.4	2.9	2.3	10.7	1.5	0.0
2016	1.8	15.9	8.4	23.6	16.2	13.1
2017	13.7	18.3	10.7	16.8	34.5	−30.6
2018	0.2	3.2	0.0	12.6	0.9	14.4
2022	1.4	3.6	–	–	3.6	5.9

Table 9. Early-window effects per 10 points (exploratory). RI is randomisation inference.

Flood record	Outcome	Sample	Effect per 10 points (%)	Clustered p	Driscoll–Kraay p	Conley p	RI p
Sentinel-1	Harvested area	All years	−14.0	0.077	< 0.001	0.043	0.092
Sentinel-1	Harvested area	Without 2017	9.9	0.085	0.009	0.068	0.161
Sentinel-1	Harvested area	Without 2016	−9.6	0.127	< 0.001	0.104	0.059
Sentinel-1	Yield	All years	−1.7	0.117	0.067	0.096	0.761
Sentinel-1	Yield	Without 2017	6.9	0.078	0.017	0.013	0.010
Sentinel-1	Yield	Without 2016	−3.2	0.047	0.045	0.029	0.321
JRC	Harvested area	All years	−19.5	0.036	< 0.001	0.011	< 0.001
JRC	Harvested area	Without 2017	−2.5	0.455	0.486	0.455	0.651
JRC	Harvested area	Without 2016	−24.3	0.054	< 0.001	0.008	0.001
JRC	Yield	All years	0.3	0.791	0.737	0.763	0.789
JRC	Yield	Without 2017	1.4	0.364	0.448	0.384	0.411

Flood record	Outcome	Sample	Effect per 10 points (%)	Clustered p	Driscoll–Kraay p	Conley p	RI p
JRC	Yield	Without 2016	0.0	0.972	0.970	0.971	0.980

5.7 Monsoon floods

Monsoon floods gave no clear result (Table 10). JRC pointed to more area (5.7 % per 10 points) and Sentinel-1 to less (−4.9 %), and neither was significant. With JRC, flooded years in every bin had higher yields than years below 1 % (Fig. 6a), consistent with earlier work (Banerjee, 2010; Asada et al., 2005), but Sentinel-1 showed nothing similar. The expected harm from extreme monsoon floods did not appear in either record. None of these tests survived correction.

Table 10. Monsoon-flood effects per 10 points of district area.

Flood record	Outcome	Effect per 10 points (%)	Clustered p	Driscoll–Kraay p	Conley p	Randomisation p
Sentinel-1	Harvested area	−4.9	0.350	0.169	0.306	0.354
Sentinel-1	Yield	0.8	0.406	0.247	0.423	0.572
Sentinel-1	Production	−4.1	0.459	0.303	0.422	0.441
JRC	Harvested area	5.7	0.073	0.098	0.054	0.090
JRC	Yield	0.4	0.690	0.616	0.649	0.655
JRC	Production	6.1	0.064	0.103	0.053	0.074

5.8 NDVI

NDVI tracked differences between districts and the shared rise in yield, but not year-to-year departures within a district (Table 11). Over January–April, the correlation with yield was 0.46 between districts and 0.42 within districts, and fell to −0.07 once year effects were removed. No window gave a significant effect, including peak greenness (Mosleh et al., 2016), and the result did not depend on how much of a district is rice land (0.05 and 0.06). NDVI-based yield estimates at national or division level (Mosleh et al., 2016; Alam et al., 2019) may therefore capture broad differences and trends, not shocks to a district.

Table 11. NDVI and Boro yield.

NDVI window	Between districts	Within districts	Within, year effects removed	Effect on log yield per 0.1 NDVI	Clustered p	Randomisation p
January-April	0.46	0.42	−0.07	−0.027	0.289	0.214

NDVI window	Between districts	Within districts	Within, year effects removed	Effect on log yield per 0.1 NDVI	Clustered p	Randomisation p
February to mid-March	0.52	0.29	0.05	0.016	0.095	0.181
Late March to early April	0.15	0.33	0.04	0.012	0.217	0.263

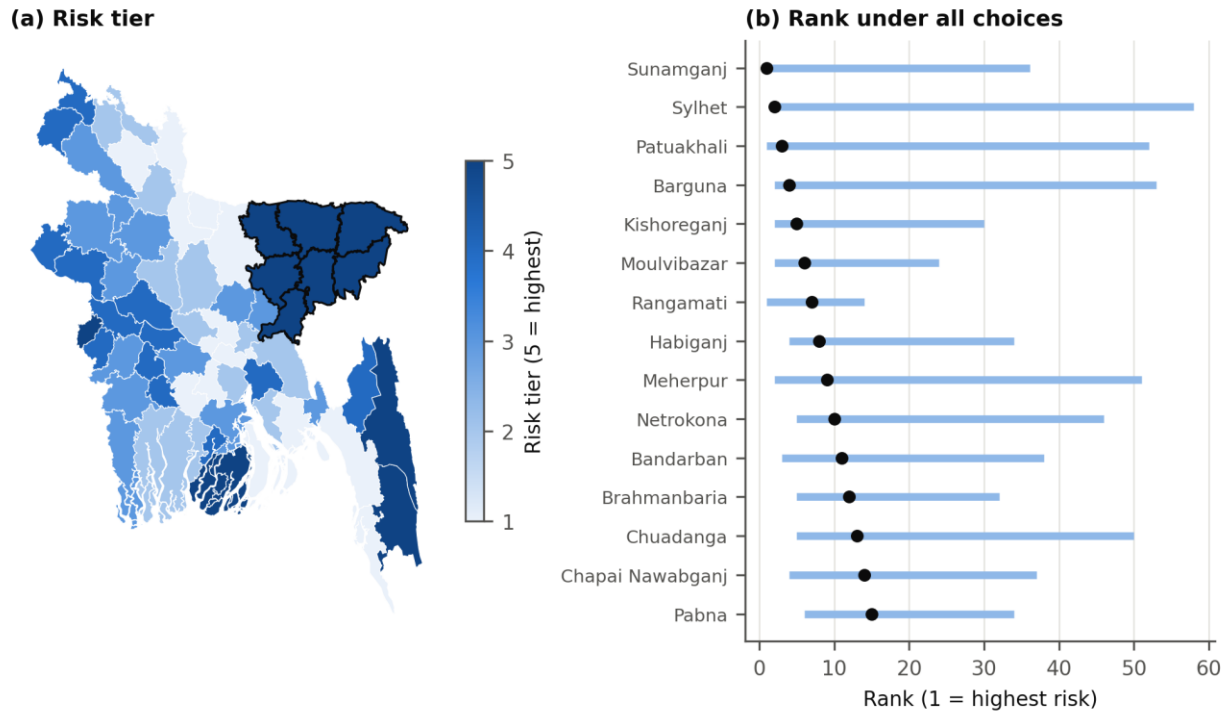
5.9 Risk index and spatial patterns

The haor dominates the index: 6 of the top ten districts are in the haor, led by Sunamganj (Table 12, Fig. 7). The ranking is not stable, however. Across 6,018 combinations, the median correlation with the base ranking was 0.66, the median district stayed in its risk tier in 35 % of cases, and no district was in the top ten in at least 90 % of them. Two results need care. Patuakhali and Barguna rank high because Boro expanded there from a very small base, which appears as instability but is not loss. Hill districts such as Rangamati have so little rice land that their flood shares are unreliable.

Table 12. The ten highest-ranked districts, with rank range and stability across all combinations.

Rank	District	Haor	Rank range, 5th–95th percentile	In top ten (%)	Same tier (%)
1	Sunamganj	Yes	1–36	88	89
2	Sylhet	Yes	2–58	77	78
3	Patuakhali	No	1–52	59	65
4	Barguna	No	2–53	56	61
5	Kishoreganj	Yes	2–30	78	81
6	Moulvibazar	Yes	2–24	88	89
7	Rangamati	No	1–14	86	93
8	Habiganj	Yes	4–34	67	72
9	Meherpur	No	2–51	34	42
10	Netrokona	Yes	5–46	60	65

Fig. 7. Risk tier and rank under all combinations. Bars show the 5th to 95th percentile of rank, and dots the base ranking.



Spatial clustering was clear (Table 13). Moran's I ranged from 0.31 to 0.45, and Habiganj, Moulvibazar, Netrokona, Sunamganj and Sylhet formed a high-risk cluster under all four neighbour definitions.

Table 13. Spatial clustering of the index under four neighbour definitions.

Neighbours	Moran's I	p	High-risk cluster, p < 0.05	After false discovery rate
Contiguity	0.451	< 0.001	Habiganj, Kishoreganj, Moulvibazar, Netrokona, Sunamganj, Sylhet	Habiganj, Sunamganj, Sylhet
4 nearest	0.419	< 0.001	Habiganj, Moulvibazar, Netrokona, Sunamganj, Sylhet	Habiganj, Moulvibazar, Sunamganj, Sylhet
6 nearest	0.313	< 0.001	Habiganj, Moulvibazar, Netrokona, Sunamganj, Sylhet	Habiganj, Moulvibazar, Sunamganj, Sylhet
100 km band	0.323	< 0.001	Habiganj, Moulvibazar, Netrokona, Sunamganj, Sylhet	Habiganj, Moulvibazar, Sunamganj, Sylhet

A yield-only view gives a different picture. Sunamganj ranks 25th of 64 on yield instability but 3rd on production instability, and the two rankings correlate at only 0.61.

6. Discussion

6.1 What can and cannot be claimed

One claim is well supported as a description of a single event. In the 2017 flood, the loss in BBS statistics appeared in harvested area and production, and almost none in yield. Three designs agree on this, in both flood records. A satellite measure that uses no BBS data places the loss in the same districts, and published damage figures are of the same order.

The broader claim, that flash floods generally reduce Boro through area, is not established. Under a rule fixed in advance it is only suggestive. The estimate rests on 2016 and 2017, the 2016 and 2022 floods left no fall in area, and the two flood records differ about five-fold in their main estimates. We report that plainly because a single significant estimate of about 22 % per 10 points would have told a stronger and less accurate story. This also bears on how flood effects are read elsewhere: with few events and a few treated districts, an estimate from a large panel can be one event in disguise.

6.2 Why area and not yield

The likely reason is how the statistics are built. Yield is estimated from crop cuts on harvested fields (Bangladesh Bureau of Statistics, 2026), so a destroyed field leaves the harvested area and never enters the yield estimate. Since Boro is planted before flash floods arrive, a same-season fall in harvested area means crop lost and not a planting decision. Crops that survive are measured as usual, which fits a yield effect near zero. The small positive yield effect with JRC could mean that weaker fields are lost first, but the evidence for this is thin.

This does not contradict Li et al. (2025), who report yield losses from submergence worldwide. Li et al. and this study measure different things in different statistical systems. Whether crop that is totally lost appears in yield or area depends on how each country's data are collected. Similarly, national data may fail to show flood effects at all (Lesk et al., 2016), because a loss such as 2017 is confined to a few districts. Our results differ from the Kansas wheat findings (Obembe et al., 2021) and resemble those on harvested ratios (Cui, 2020), which suggests that the answer depends on the crop, hazard and data system.

6.3 Implications

Disaster assessment. Yield alone is a poor signal of the largest flood losses in this system. Sunamganj lost 73 % of its production in 2017 with no visible change in yield. Assessments should track harvested area and production alongside yield, and research on Bangladesh's rice should not treat stable yields as evidence of low flood risk. Harvested area is the variable that carried the signal here, but BBS reports

only harvested and not planted area. A routine estimate of planted area before the flood window would show directly what share of the crop was lost.

Satellite monitoring. Greenness-based crop loss located the 2017 damage without official data, as earlier work found for the same event (Ahmed et al., 2017), and similar methods have been used elsewhere (Rahman and Di, 2020; Qamer et al., 2023). It is useful for rapid assessment, but the 2016 mismatch shows that it should not be read as loss without information on harvest progress. NDVI alone should not be used to infer yield shocks in a district.

Timing and adaptation. If water before harvest is what matters, then the most direct response is to finish harvesting before the flood window. Earlier sowing and short-duration varieties are already recommended for this reason (Hossain et al., 2023), though farmers face barriers to adopting them (Baishakhy et al., 2023). The exploratory timing result supports this direction but does not establish it. Flood-tolerant rice has improved outcomes in north-west Bangladesh (Bairagi et al., 2021), but that evidence does not transfer directly to Boro in the haor, where the threat comes at maturity. Interest in crop insurance exists (Islam et al., 2021), and our results suggest that a payout tied to district yield would not have been triggered in 2017. Any index for the haor should be based on area or on satellite-observed loss.

6.4 Limitations

The study covers 15 years and few large flash floods, and one event dominates, so the findings describe these events and do not give a general law. Floods are not randomly assigned. BBS harvested area comes from official estimates, which we checked with independent evidence only partly, and only for large events. The Boro mask relies on a fixed NDVI rule and a static cropland map, is not validated with field data, and has noisy yearly totals. Sentinel-1 has fewer scenes before 2017, and optical records such as JRC can miss short floods that radar observes (Uddin et al., 2019; Twele et al., 2016). Flood fractions in the main models cover whole districts, and the required rice-only check weakened the estimates. Satellite crop loss cannot separate destruction from early harvest. Cold-stress thresholds are provisional. The risk index omits adaptive capacity and salinity (Dasgupta et al., 2018), so it is a partial picture, and its rank ranges are wide. It should guide where to look more closely, and should not be used as an allocation rule.

7. Conclusion

Using printed BBS statistics for all 64 districts and an analysis plan fixed in advance, we asked whether flash floods reduce Boro rice through harvested area rather than yield. For the 2017 flood the answer is clearly yes: haor area and production fell by about a third, yield barely moved, and independent evidence

supports the loss. As a general rule the answer is only suggestive, because the estimates rest on 2016 and 2017, and other large floods left no fall in area. Flood timing is the likely reason, but that evidence is exploratory and rests on one event. The practical lesson is that yield alone can miss the largest flood losses in this system, so flood assessment and risk indices should also use harvested area and production.

CRediT authorship contribution statement

Tasnim Ahmad Mumu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. Rezaul Karim: Supervision, Methodology, Validation, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with code development, data validation and analysis scripts, and drafting and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The panel is archived at https://doi.org/10.5281/zenodo.23795046. The derived data remain subject to the terms of their sources. The BBS yearbook PDFs and the GADM v4.1 boundaries (GADM, 2022) are not redistributed and must be obtained from the Bangladesh Bureau of Statistics and gadm.org. Code,

notebooks, the analysis plan and the Earth Engine extracts for this paper are in the repository given in the Software and data availability section.

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Appendix A. Supplementary tables

Table A.1. Other variants of the area effect, reported without a pass rule. The Poisson model (Silva and Tenreyro, 2006) gives more weight to large districts.

Variant	Flood record	Area effect (%)	Clustered p	Driscoll–Kraay p
Division-by-year effects	Sentinel-1	−20.4	0.012	< 0.001
Division-by-year effects	JRC	−6.6	0.003	0.042
District linear trends	Sentinel-1	−15.7	0.002	0.018
District linear trends	JRC	−5.0	0.107	0.162
Poisson model	Sentinel-1	−10.9	< 0.001	–
Poisson model	JRC	−5.3	< 0.001	–
Flash flood only, 2015–2025	Sentinel-1	−20.7	0.018	< 0.001
Water threshold −16 dB	Sentinel-1	−20.0	0.035	< 0.001
Water threshold −20 dB	Sentinel-1	−28.8	0.001	< 0.001
Neighbours' flood, own effect	Sentinel-1	−21.5	0.049	< 0.001
Neighbours' flood, own effect	JRC	−6.2	0.013	0.052
Neighbours' flood, spillover	Sentinel-1	−0.3	0.972	0.908
Neighbours' flood, spillover	JRC	3.3	0.257	0.386

Table A.2. National area of the yearly Boro mask, against BBS and published maps.

Harvest year	Yearly mask (million ha)	BBS harvested area (million ha)	Shew and Ghosh (2019) (million ha)
2011	3.42	4.77	–
2012	3.37	4.81	–
2013	3.83	4.76	–
2014	3.79	4.79	4.29
2015	4.24	4.84	4.42
2016	4.16	4.77	4.64
2017	4.12	4.48	4.12
2018	3.65	4.86	4.41
2019	4.47	4.79	–
2020	4.18	4.76	–
2021	4.17	4.79	–
2022	3.73	4.82	–

Harvest year	Yearly mask (million ha)	BBS harvested area (million ha)	Shew and Ghosh (2019) (million ha)
2023	3.95	4.85	–
2024	3.36	4.88	–
2025	4.45	4.89	–

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